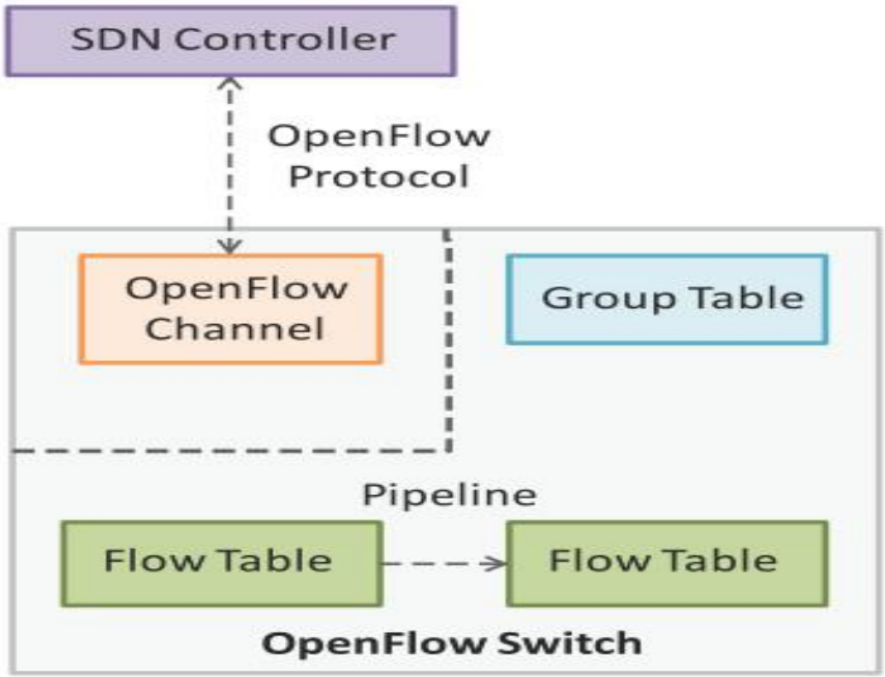
# IT Workshop II (Cloud Computing): Services and Platforms

August 24, 2026

# OpenFlow Switch



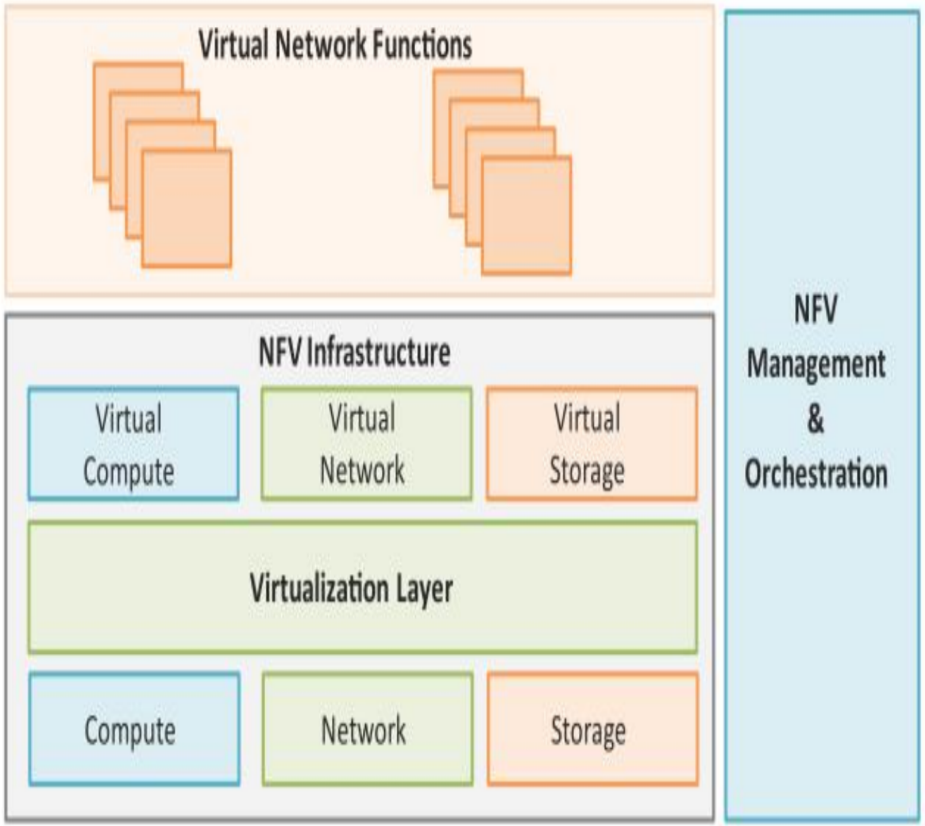
# Network Function Virtualization

- Network Function Virtualization (NFV) is a technology that leverages virtualization to consolidate the heterogeneous network devices onto industry standard high volume servers, switches and storage.
- **Relationship to SDN:** NFV is complementary to SDN as NFV can provide the infrastructure on which SDN can run. NFV and SDN are mutually beneficial to each other but not dependent.
- NFV comprises of network functions implemented in software that run on virtualized resources in the cloud.
- NFV enables a separation the network functions which are implemented in software from the underlying hardware.

# NFV Architecture

- **Virtualized Network Function (VNF):** VNF is a software implementation of a network function which is capable of running over the NFV Infrastructure (NFVI).
- **NFV Infrastructure (NFVI):** NFVI includes compute, network and storage resources that are virtualized.
- **NFV Management and Orchestration:** NFV Management and Orchestration focuses on all virtualization-specific management tasks and covers the orchestration and lifecycle management of physical and/or software resources that support the infrastructure virtualization, and the lifecycle management of VNFs.

# NFV Architecture



# Amazon DynamoDB

- Amazon DynamoDB is the non-relational (No-SQL) database service from Amazon.
- The DynamoDB data model includes include tables, items and attributes.
- A table is a collection of items and each item is a collection of attributes.
- To store data in DynamoDB you have to create a one or more tables and specify how much throughput capacity you want to provision and reserve for reads and writes.

# Application Runtimes and Framework

- Cloud-based application runtimes and frameworks allow developers to develop and host applications in the cloud.
- Application runtimes provide support for programming languages (e.g., Java, Python, or Ruby).
- Application runtimes automatically allocate resources for applications and handle the application scaling, without the need to run and maintain servers.

# Google App Engine

- Google App Engine is the platform-as-a-service (PaaS) from Google, which includes both an application runtime and web frameworks.
- App engine features include runtimes, secure sandbox environment, Python web application framework, no-SQL data storage, authentication, email service, task queues etc.

# Queueing Services

- Queueing services allow de-coupling application components.
- The de-coupled components communicate via messaging queues.
- Queues are useful for asynchronous processing.
- Another use of queue is to act as overflow buffers to handle temporary volume spikes.

# Amazon Simple Queue Service

- Amazon Simple Queue Service (SQS) is a distributed queue that supports messages of up to 256 KB in size.
- SQS supports multiple writers and readers and locks messages while they are being processed.
- SQS does not guarantee that messages will be delivered in FIFO order.
- Applications that require FIFO ordering of messages can place additional sequencing information in each message so that they can be re-ordered after retrieving from a queue.

# Amazon Simple Email Service

- Amazon Simple Email Service (SES) is bulk and transactional email-sending service from Amazon.
- SES is an outbound-only email-sending service that allows applications hosted in the Amazon cloud to send emails.
- SES uses content filtering technologies to scan the outgoing email messages.
- SES service can be accessed and used from the SES console, the Simple Mail Transfer Protocol (SMTP) interface, or the SES API.

# The End